

Name: Ameer Edzzad Shah

Student ID: 22005680

Group: G8

The original Lab 3 exercises were built around the Tips dataset, which centers on restaurant transactions and customer tipping patterns. For this assignment, that dataset was swapped out for the Superstore Sales dataset in order to examine business performance instead. As a result, the focus of the visualizations moved away from customer behavior and toward sales figures, profit margins, product categories, and regional performance trends. Because the Superstore dataset includes a wider range of business-related variables, it allowed for more layered, multidimensional analysis. Although the same visualization methods were applied, the insights generated were now more directly applicable to business decision-making and sales strategy. Additionally, the broader spread of numerical values in the Superstore dataset made patterns, clusters, and outliers stand out more clearly across multiple charts.

Modification 1

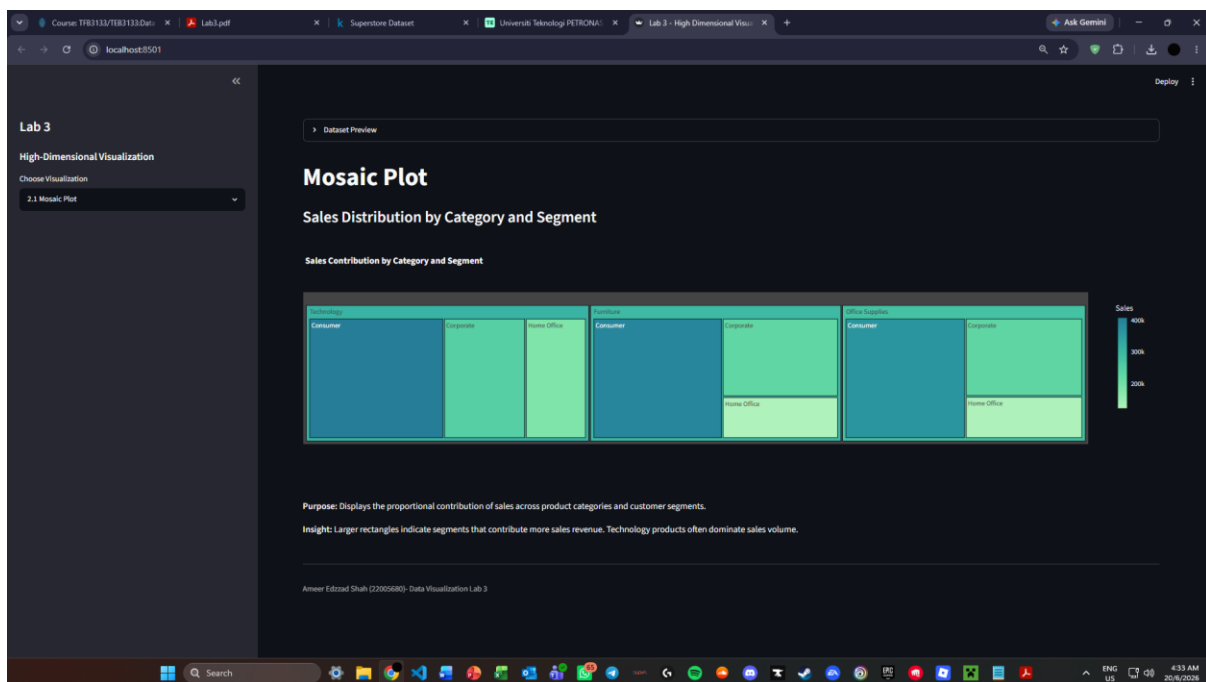
Replaced restaurant-related variables with business sales variables.

Modification 2

Updated the color scheme.

Modification 3

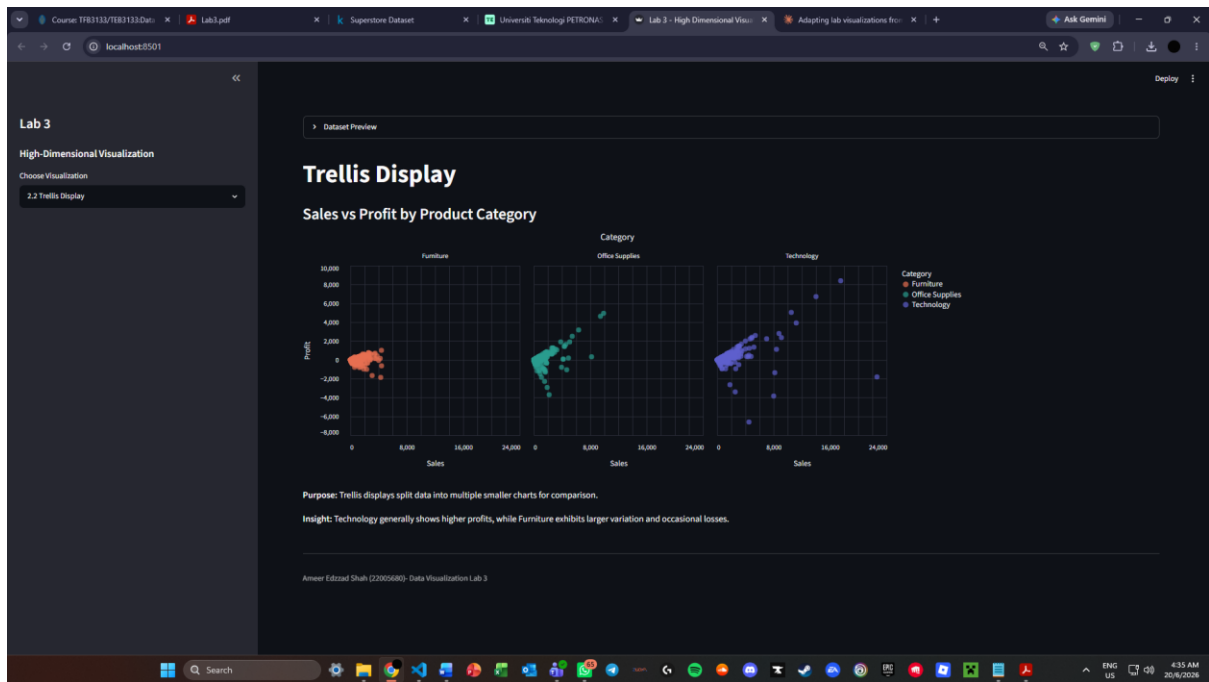
Enhanced titles and tooltips for clarity.



Visualization 1: Mosaic Plot

Purpose: Illustrates the proportional contribution of sales across different categories and customer segments.

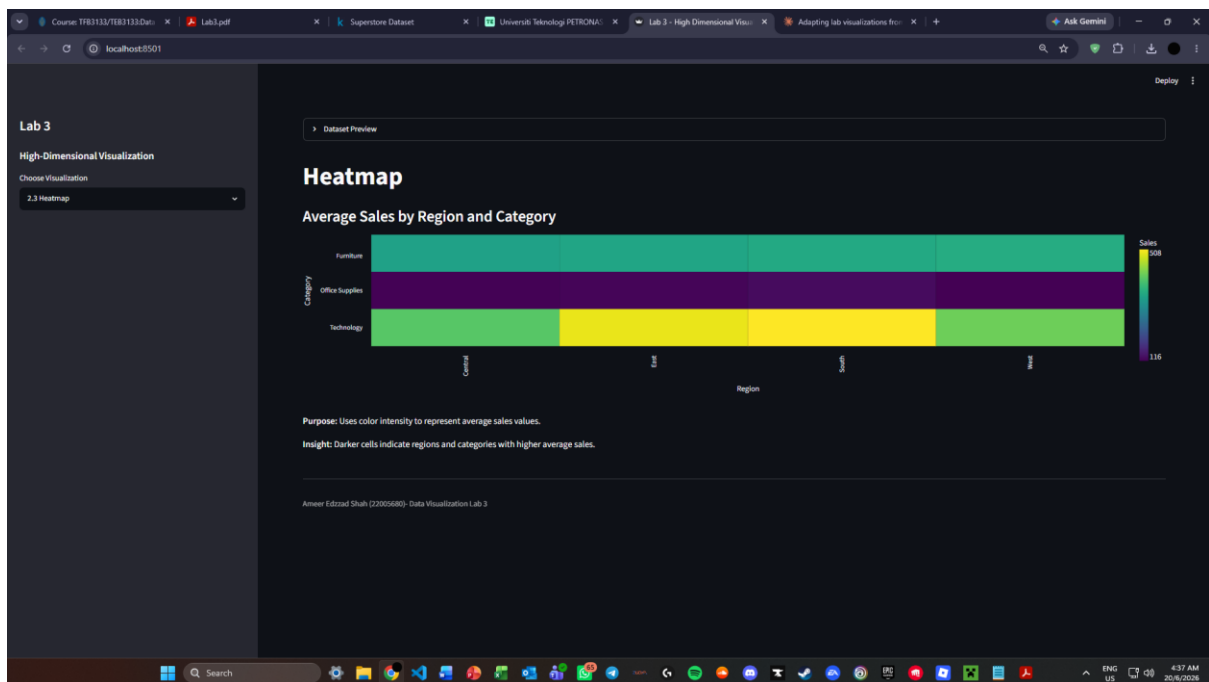
Insight: A significant portion of Technology sales comes from Consumer-segment customers.



Visualization 2: Trellis Display

Purpose: Examines the relationship between sales and profit across different product categories.

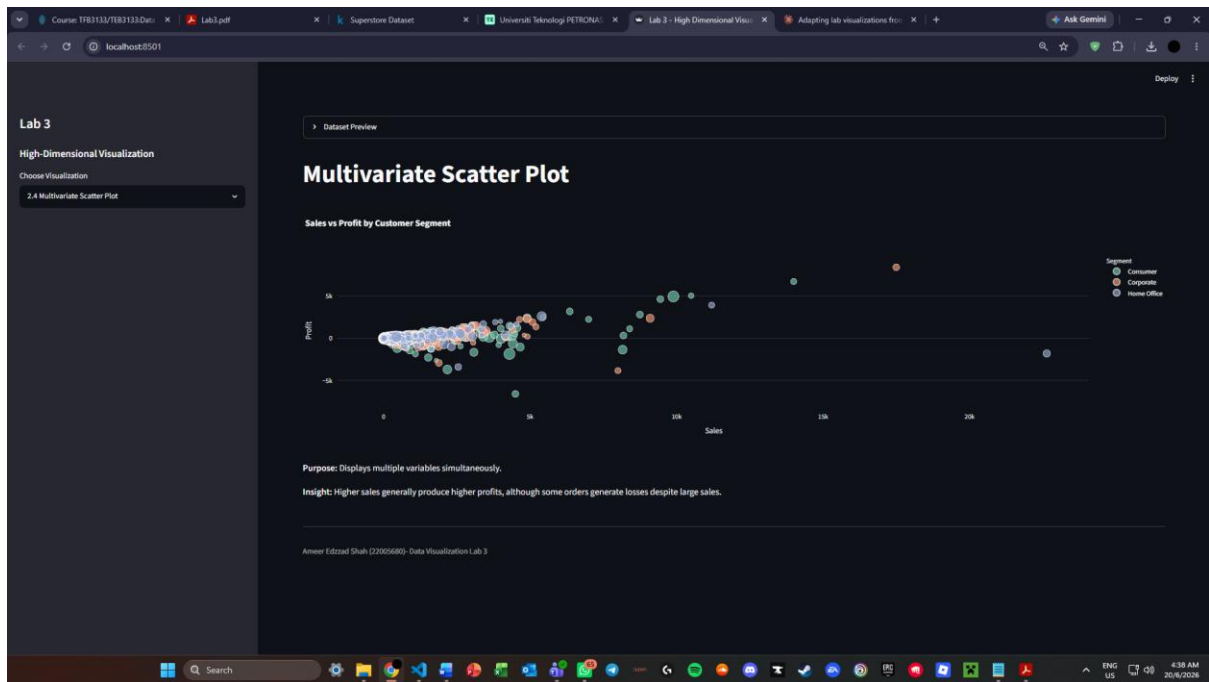
Insight: Technology tends to generate higher profit levels than Furniture.



Visualization 3: Heatmap

Purpose: Shows average sales intensity.

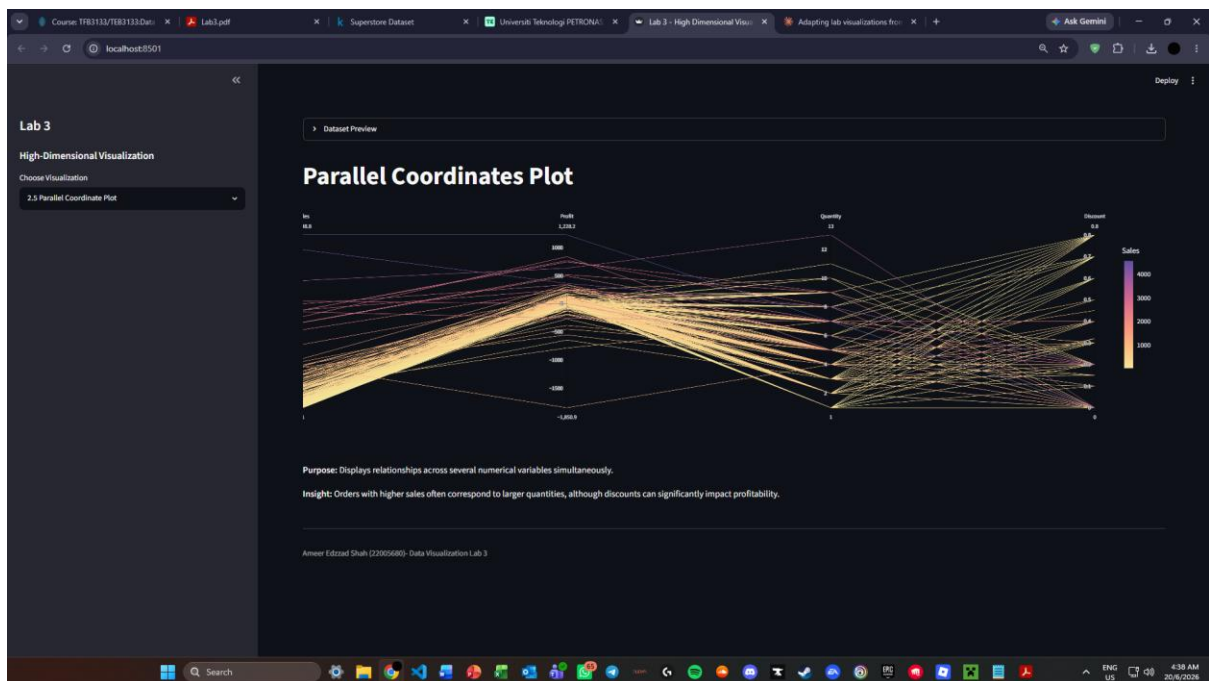
Insight: The West region records higher average sales.



Visualization 4: Multivariate Scatter Plot

Purpose: Displays multiple variables simultaneously.

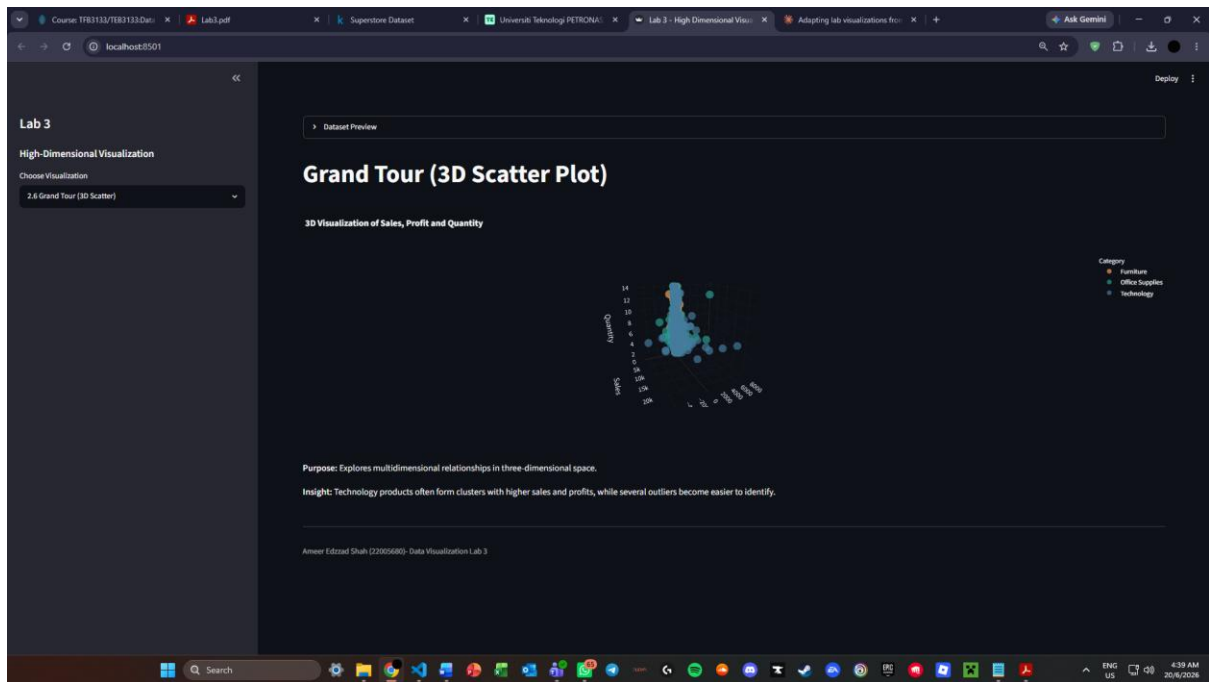
Insight: Higher sales generally correspond to higher profits, although exceptions exist.



Visualization 5: Parallel Coordinate Plot

Purpose: Explores multidimensional relationships.

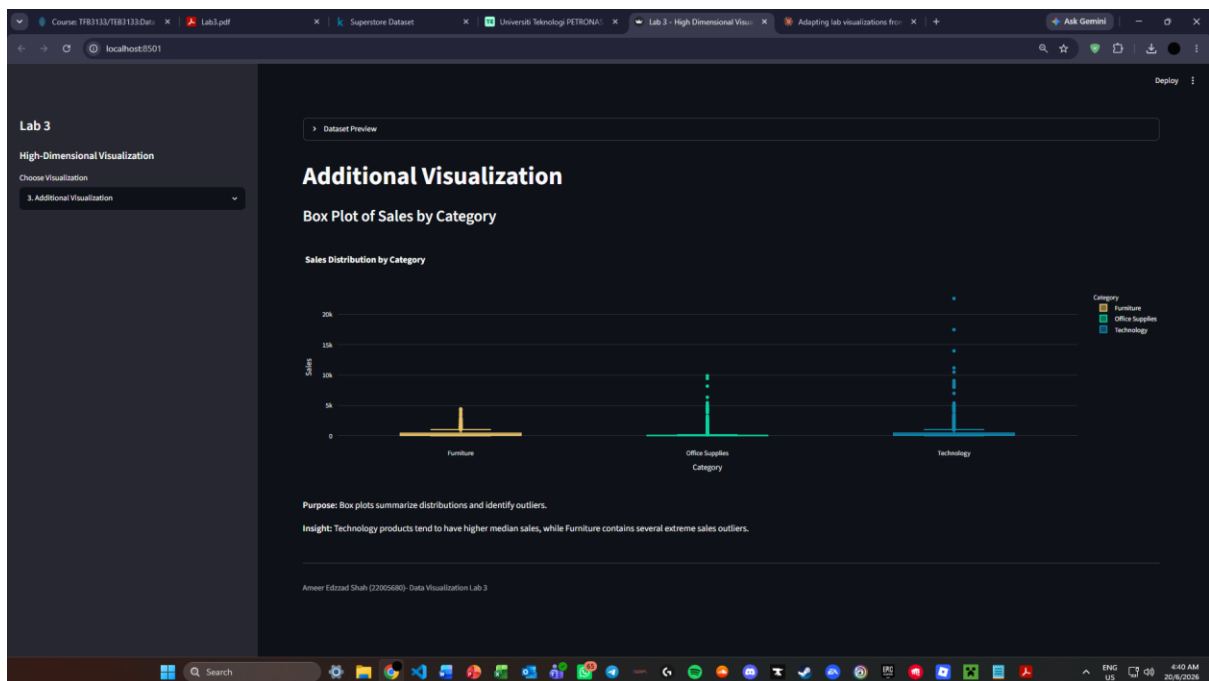
Insight: Large sales orders tend to involve larger quantities.



Visualization 6: Grand Tour (3D Scatter)

Purpose: Visualizes data in three-dimensional space.

Insight: Several high-sales outliers become easier to identify.



Visualization 7: Box Plot

Purpose: Shows distribution and outliers.

Insight: Technology products have higher median sales.